Question 1

Identify the network interfaces currently active on the system.

Approach

I used a networking command to display details of all available network interfaces and their configurations.

Commands

- `ip a` → displays all network interfaces
  OR
- `ifconfig` → shows network interface configuration

Output

STEP 1 : Show network interfaces

```
┌──(aman㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:7b:ec:7b brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 83707sec preferred_lft 83707sec
    inet6 fd17:625c:f037:2:271e:efe2:ad2f:8fec/64 scope global temporary dynamic
       valid_lft 86002sec preferred_lft 14002sec
    inet6 fd17:625c:f037:2:a00:27ff:fe7b:ec7b/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86002sec preferred_lft 14002sec
    inet6 fe80::a00:27ff:fe7b:ec7b/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Key Observations

Interfaces such as :

- eth0 → wired interface
- lo → loopback interface

What I learned

Linux systems use network interfaces to communicate with other systems and networks.

Question 2

Assign a new IP address to the eth0 network interface.

Approach

I manually configured the IP address for the eth0 interface.

Commands

- `ifconfig` → configure interface
  OR modern command:
- `ip addr add` → assigns IP address

Output

STEP 1 : Assign IP address

```
┌──(aman㉿kali)-[~]
└─$ sudo ip addr add 192.168.1.1/24 dev eth0
```

STEP 2 : Verify IP change

```
┌──(aman㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:7b:ec:7b brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 83279sec preferred_lft 83279sec
    inet 192.168.1.1/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fd17:625c:f037:2:271e:efe2:ad2f:8fec/64 scope global temporary dynamic
       valid_lft 86056sec preferred_lft 14056sec
    inet6 fd17:625c:f037:2:a00:27ff:fe7b:ec7b/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86056sec preferred_lft 14056sec
    inet6 fe80::a00:27ff:fe7b:ec7b/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Key Observations

The IP address for eth0 changes to the assigned value.

What I learned

IP addresses identify devices in a network and can be manually configured.

Question 3

Modify the MAC address of the eth0 network interface.

Approach

I changed the hardware address (MAC address) associated with the network interface.

Commands

- `ifconfig` → changes MAC address
  OR
- `ip link` → modern command

Output

STEP 1 : Disable interface

```
┌──(aman㉿kali)-[~]
└─$ sudo ifconfig eth0 down
[sudo] password for aman:
```

STEP 2 : Change MAC address

```
┌──(aman㉿kali)-[~]
└─$ sudo ifconfig eth0 hw ether 00:11:22:33:44:55
```

STEP 3 : Enable interface

```
┌──(aman㉿kali)-[~]
└─$ sudo ifconfig eth0 up
```

STEP 4 : Verify change

```
┌──(aman㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:11:22:33:44:55 brd ff:ff:ff:ff:ff:ff permaddr 08:00:27:7b:ec:7b
    inet 10.0.2.16/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86391sec preferred_lft 86391sec
    inet6 fd17:625c:f037:2:a4cf:a719:c29e:89ee/64 scope global temporary dynamic
       valid_lft 86392sec preferred_lft 14392sec
    inet6 fd17:625c:f037:2:211:22ff:fe33:4455/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86392sec preferred_lft 14392sec
    inet6 fe80::211:22ff:fe33:4455/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Key Observations

MAC address is updated to the new value.

What I learned

MAC addresses uniquely identify network devices and can be modified for testing or privacy.

Question 4

Check if any wireless network interfaces are available on the system.

Approach

I used a command that lists wireless network interfaces.

Commands

- `iwconfig` → shows wireless interfaces

Output

STEP 1 : Check wireless interfaces

```
┌──(aman㉿kali)-[~]
└─$ iwconfig
lo        no wireless extensions.

eth0      no wireless extensions.
```

Key Observations

Wireless interfaces typically appear as wlan0 or similar.

What I learned

Wireless interfaces allow connection to WiFi networks.

Question 5

Obtain a new IP address automatically from a DHCP server.

Approach

I requested the system to obtain an IP address dynamically from the network.

Commands

- `dhclient` → requests IP from DHCP server

Output

STEP 1 : Request DHCP IP

```
┌──(aman㉿kali)-[~]
└─$ sudo dhclient -r eth0
Killed old client process

┌──(aman㉿kali)-[~]
└─$ sudo dhclient eth0
```

STEP 2 : Verify new IP

```
┌──(aman㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:11:22:33:44:55 brd ff:ff:ff:ff:ff:ff permaddr 08:00:27:7b:ec:7b
    inet 10.0.2.16/24 brd 10.0.2.255 scope global dynamic eth0
       valid_lft 84881sec preferred_lft 84881sec
    inet6 fd17:625c:f037:2:a4cf:a719:c29e:89ee/64 scope global temporary dynamic
       valid_lft 86113sec preferred_lft 14113sec
    inet6 fd17:625c:f037:2:211:22ff:fe33:4455/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86113sec preferred_lft 14113sec
    inet6 fe80::211:22ff:fe33:4455/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Key Observation

DHCP automatically assigns IP configuration.

What I learned

DHCP simplifies network configuration by automatically assigning IP addresses.

Question 6

Identify the DNS server and mail associated with a website.

Approach

I queried DNS records to find nameserver and mail server details.

Commands

- `nslookup` → queries DNS
- `dig` → advanced DNS lookup

Output

STEP 1 : Find nameserver

```
┌──(aman㉿kali)-[~]
└─$ nslookup google.com
Server:         10.233.189.22
Address:        10.233.189.22#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.24.78
Name:   google.com
Address: 2404:6800:4002:823::200e
```

STEP 2 : Find mail server

```
┌──(aman㉿kali)-[~]
└─$ nslookup -type=mx google.com
Server:         10.233.189.22
Address:        10.233.189.22#53

Non-authoritative answer:
google.com      mail exchanger = 10 smtp.google.com.

Authoritative answers can be found from:
```

Key Observations

DNS records include:

- NS → nameserver
- MX → mail server

What I learned

DNS records provide information about services associated with a domain.

Question 7

Configure the system to use Google's DNS server for domain name resolution.

Approach

I edited the DNS configuration file to include Google's public DNS server.

Commands

- `nano` → edit configuration file
- `8.8.8.8` → Google DNS server

Output

STEP 1 : Open DNS config file

STEP 2 : Add Google DNS entry

STEP 3 : Save file

Key Observations

System will use Google DNS if local DNS fails.

What I learned

DNS configuration affects how domain names are translated into IP addresses.

Security Insights

- Attackers often gather information about network interfaces
- IP addresses help identify devices in a network
- MAC address changes can help anonymize devices
- DNS information reveals infrastructure details about websites
- DHCP and DNS manipulation can be used in network attacks